Chase Oswalt

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EDUCATION

University of Central Florida - Orlando, FL
Bachelor of Science in Information Technology | Graduated: August 2026

- **GPA:** 3.53/4.00 | **Honors:** Dean's List
- **Awards:** Elbit America STEM Scholarship, Florida Bright Futures Medallion Scholarship
- **Coursework:** Systems Software, Object-Oriented Programming, Enterprise Computing, Database Systems, System Administration & Maintenance, Human-Computer Interaction

TECHNICAL SKILLS

- **Languages:** C (C99), C++, C#, Java, Python, JavaScript (ES6+), SQL, PowerShell, HTML5/CSS3, GLSL
- **Embedded & Low-Level:** Bare-Metal Firmware, AVR (ATmega328P), Register-Level Drivers (UART/USART, I2C/TWI, SPI, Timers, ADC), Hardware Interrupts (ISRs), Static Memory Management
- **Web & Frameworks:** Docker, Docker compose, React.js, Node.js, Express.js, Java Servlets, JSP, JDBC, REST APIs, Vite, Nginx
- **Systems, Cloud & Tools:** AWS (Cloud Architecting Certified), Linux, Windows Server 2022, Active Directory, DigitalOcean, Docker, VirtualBox, Git, GitHub Actions (CI/CD), MongoDB, MySQL, Jest, Postman

TECHNICAL PROJECTS

Bare-Metal Ground Launch Sequencer (GLS) Controller | *C (C99), ATmega328P, avr-gcc, GNU Make, USART, I2C*

- Developed bare-metal launch sequencer firmware in C99 on an 8-bit ATmega328P (16 MHz) without Arduino libraries or external HALs.
- Wrote register-level drivers from scratch for USART (9600-baud serial telemetry), hardware I2C/TWI (OLED status display), and Timer1 in CTC mode to generate a deterministic 1 ms system tick.
- Implemented a non-preemptive cyclic executive scheduler and Finite State Machine (FSM) managing Power-On Self-Test (POST), safety interlocks, T-minus countdown, and abort states.
- Built a 4x4 matrix keypad scanner using direct port manipulation (DDRx, PORTx, PINx) and timer-based debouncing for dual-PIN operator authentication.
- Prevented heap fragmentation and race conditions by eliminating dynamic allocation (malloc), using volatile-qualified static ring buffers between ISRs and the main loop.

UCF Technology Lending Web Platform | *React, Node.js, Express, MongoDB, DigitalOcean, GitHub Actions*

- Led a 4-person Agile team to build a full-stack MERN application replacing a legacy university equipment checkout system across Student, Faculty, and Admin roles.
- Containerized a full-stack MERN application using **Docker** and **Docker Compose**, implementing a multi-stage Node/Nginx build for the Vite frontend and isolating client/server services across a custom bridge network.
- Implemented authentication and security using Google OAuth 2.0, JWT middleware, bcrypt password hashing, and the Resend API for automated email verification.
- Integrated ZXing webcam barcode scanning and external REST APIs to speed up hardware cataloging and check-in/check-out workflows.
- Configured a GitHub Actions CI/CD pipeline with automated Jest/Supertest suites deploying to a DigitalOcean Droplet behind Nginx (HTTP/2, Gzip, static caching), achieving a >95% Lighthouse performance score.

Multi-Role Database Web Application | *Java, JSP, Servlets, Apache Tomcat, MySQL, JDBC*

- Built a three-tier Java web application on Apache Tomcat backed by a MySQL database to manage role-restricted business transactions and audit logs.
- Enforced Role-Based Access Control (RBAC) across four permission tiers (Root, Client, Data Entry, Accountant) with distinct SQL connection profiles.
- Wrote server-side business logic using JDBC PreparedStatement and CallableStatement to execute stored procedures and prevent SQL injection.

Windows Server Active Directory & Automation Lab | *Windows Server 2022, Active Directory, PowerShell, Python*

- Configured a multi-VM enterprise network in Oracle VirtualBox featuring Windows Server 2022 domain controllers and client workstations.
- Set up Active Directory Domain Services (AD DS), Organizational Units (OUs), Group Policy Objects (GPOs), and DNS/DHCP services for centralized access control.
- Wrote PowerShell and Python scripts to automate bulk user provisioning, system health checks, and routine maintenance.

2D Physics Arena Combat Game | *Godot Engine, C#, GLSL* | *youtu.be/4_mdrE4RQnI*

- Programmed core state-machine combat logic, collision handling, and local multi-controller input in C# within the Godot Engine.
  Wrote custom GLSL fragment shaders for real-time lighting and post-processing effects, along with a CSV data loader for runtime character balancing.

EXTRACURRICULAR INVOLVEMENT

AI@UCF & GameDev Knights - Member: Participated in collaborative game jams and technical workshops covering neural network fine-tuning, game design patterns, and physics engines.